| NOT MEASUREMENT |
| SENSITIVE |

MIL-C-23565E
14 January 1989
SUPERSEDING
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5 March 1982

MILITARY SPECIFICATION

COUNTER, DISH-TRAY CONVEYOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dish-tray conveyor counters.

1.2 Classification. The dish-tray conveyor counters are of the following types, classes, and models, as specified (see 6.2):

Type I - Stationary.
Type II - Mobile.

Class 1 - Without belt washer.
Class 2 - With belt washer.

Model a - Without garbage disposal unit, silver soak sink, and scrapping trough.
Model b - With garbage disposal unit, silver soak sink, and scrapping trough.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7320

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and drawings. The following specifications, standards, and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronics and Associated Equipment).
MIL-F-3541 - Fittings, Lubrication.
MIL-G-15840 - Garbage Disposal Machines, Commercial and Household.
MIL-K-43875 - Kitchen Equipment, Including Unit Assemblies, Repair Parts and Tools, Preparation for Delivery of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

DRAWING

NAVY FOOD SERVICE SYSTEMS OFFICE

Project 11103, File 541

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue



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of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI B20.1 - Safety Standard for Conveyors and Related Equipment.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3308.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 167 - Standard Specification for Stainless and Heat-Resisting Chromium Nickel Steel Plate, Sheet and Strip. (DoD adopted)
ASTM A 176 - Standard Specification for Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip. (DoD adopted)
ASTM A 480 - Standard Specification for General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 3 - Industrial Systems.
NEMA ICS 6 - Enclosures for Industrial Controls and Systems.
NEMA MG 1 - Motors and Generators. (DoD adopted)
NEMA WD 1 - Wiring Devices, General Requirements for.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

NATIONAL SANITATION FOUNDATION (NSF)

NSF No. 2 - Food Service Equipment.
Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, 3475 Plymouth Road, P. O. Box 1468, Ann Arbor, MI 48106.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These



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documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The dish-tray conveyor counter shall be of the open pan-type or enclosed pan type when specified (see 6.2), and shall automatically move food trays, dishes, glassware, and silverware from the loading end to the discharge end of the unit. The conveyor counter shall consist essentially of a belt pan having head- and tail-end sections, motor, belt, motor control, automatic belt take-up where required, electrical wiring, and other components and accessories specified herein to form an integral unit.

3.2 First article. When specified (see 6.2), the contractor shall furnish a conveyor counter of the type, class, and model as required for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The conveyor counter shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the of used or rebuilt products are allowed under this specification.

3.4.1 Corrosion-resisting steel. Corrosion-resistant steel shall be class 302 or 304 of ASTM A 167, or class 430 of ASTM A 176, annealed or tempered to suit the intended application. The surface finish shall be No. 3 or 4 as defined in ASTM A 480, except that concealed surfaces may be finished in accordance with the manufacturer's standard practice.



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3.4.2 Dissimilar metals. In fabricating the conveyor counter, the use of dissimilar metals in intimate contact with each other shall be avoided wherever possible to minimize galvanic corrosion. When use of dissimilar metals cannot be avoided, the joints or contact areas shall be protected against galvanic corrosion by plating, coating, gasketing, or other interposing material unless protection is inherently provided by area relationships of the anodic and cathodic materials. Bolts, nuts, pins, screws, and other fastenings shall be the same material as the metals joined or shall be cathodic to the metals joined.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Standards compliance. Conveyor counters shall meet the requirements of NSF No. 2.

3.6.1 Certification. As part of the first article test, if a first article is submitted, or prior to the approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the dish-tray conveyor counter he proposes to supply under this specification meets the requirements of NSF No. 2.

3.6.1.1 NSF certification. Acceptable evidence of meeting the requirements of NSF No. 2 shall be the NSF seal on the finished dish-tray conveyor counters and inclusion in the current edition of the NSF Listing of Food Service Equipment.

3.6.2 Alternate certification. In lieu of the certification specified in 3.6.1, acceptable evidence of meeting the requirements of the standard will be a certified test report from a nationally recognized independent testing laboratory, approved by the contracting officer, or acceptable to the medical department of the service for which the conveyor counters are being acquired attesting that a conveyor counter typical in all respects of the units to be supplied under this specification has been examined and tested, and meets all requirements of NSF No. 2. When authorized in writing by the contracting officer, such examination and tests for alternate certification may be conducted by the manufacturer at the manufacturer's plant when the knowledgeable inspection authority has determined that adequate facilities are available for conducting the required test.

3.6.3 Applicability. The conveyor counters shall meet the dimensional requirements, belt speed, and all other requirements of this specification. The contractor shall select the conveyor counter to be offered as meeting these requirements from his models which are NSF listed.

3.7 Performance. The dish-tray conveyor counter shall meet the performance requirements specified herein. The operation shall be completely automatic when the conveyor counter is connected to the electrical power supply, and for class 2, and model b when the unit is connected to the water supply. The



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conveyor counter shall meet the requirements of the operational test of 4.6.1 and the performance test of 4.6.3.

3.8 Design. The conveyor counter shall be designed to facilitate access for maintenance and servicing (including cleaning) of components in the field. The counter shall conform to the applicable safety requirements of ANSI B20.1 and as specified herein. Parts so located as to be hazardous to operating personnel, and all rotating or reciprocating parts, shall be fully enclosed or properly guarded. Protection to personnel shall be provided in accordance with NFPA No. 70, National Electrical Code. All screw threads shall conform to FED STD H28.

3.9 Construction. The belt conveyor counter, and component parts thereof, shall be constructed in accordance with Naval Food Service System Office Drawing Project 11103, File 541, NSF No. 2, and as specified herein. The counter shall be constructed of structural steel or square tubing frames in accordance with the manufacturer's standard practice. The frames shall be of adequate strength, braced, and welded to form corrosion-resistant rigid structures. The frames shall be covered with not less than 18-gage corrosion-resistant steel as specified in 3.4.1. The top edges of the counter shall be formed into a rolled rim flange 3 inches (76.2 millimeters (mm)) high, and having a minimum diameter of 1.250 inches (31.75 mm). The flange shall have welded, watertight, smooth rounded corners, and shall conform to NSF No. 2. The top of the flange for model a counters shall be 37 inches (939.8 mm) above the floor. The top edge at the discharge end shall be flanged downward in a suitable discharge arrangement to meet a table top 34 inches (863.6 mm) high. The model b counters shall have a $180^{o}$ rolled rim on the ends, front, and back of the conveyor. The top of the rim on the back of model b counters shall be 38 inches (965.2 mm) above the floor. The length of the counter shall be as specified (see 6.2). The length of pan and belt shall be in accordance with the manufacturer's standard practice for the counter furnished. The edges of adjoining sections shall be rolled, evenly matched, and rigidly held together with suitable splices. The ends of rolled edges on terminal sections shall be closed by welding filler plates to the counter. All bearings shall be the replaceable, sealed-for-life type. Component parts shall be complete and assembled properly. Moving parts shall have no undue vibration, bearings shall not overheat, and starting and stopping and adjusting devices shall operate in a positive, smooth, and efficient manner when the conveyor counter is tested as specified in 4.6.1.

3.9.1 Drain pan. The drain pan shall be 18 gage, corrosion-resistant steel, approximately 15-18 inches wide inside, and have a maximum overall width of 21 inches (533.4 mm). The pan shall be suspended under the full length of the conveyor belt (see 3.9 and 3.9.3) so constructed and mounted that it may be removed and replaced in the field without disassembling other components. The pan shall pitch toward the drain, and corners shall be in accordance with NSF No. 2. Pitch shall be sufficient to avoid standing water in the pan.

3.9.2 Head- and tail-end sections. The head- and tail-end sections shall be completely enclosed with 18 gage corrosion-resisting steel. The sections shall have removable access panels or hinged doors. The head- and tail-end



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sections shall be furnished with suitable threaded drain fittings. Snub- and head-pulleys shall be mounted on the structured steel frames. When a belt washing system is supplied with the conveyor, bearings may be mounted to the 14 gauge stainless steel wash tank. Where required, an automatic belt take-up device shall be furnished and installed in the head-end section. The shafting and pulleys shall be of stainless steel to prevent rust stains on the belt.

3.9.3 Belt. The conveyor belt shall be furnished in the required length for use with the pan specified (see 3.9 and 3.9.1). The width of the belt shall be 10 or 12 inches (254 or 304.8 mm), as specified (see 6.2). The belt shall be resistant to the effects of grease, food acids, or other deleterious foodstuff ingredients. The belt shall be slat- or flexible-type as specified (see 6.2). The slat-type belts shall be assembled on a stainless steel carrier chain and a full length wear strips constructed of Polyethylene Resins (UHMW) plastic to protect the conveyor top wear shall also be furnished.

3.9.4 Rollers. The conveyor counter shall be furnished with nonmarking 2-inch diameter plastic idler rollers where required to support the belt on the return run. The rollers shall be the ball-bearing type, having permanently sealed-in lubrication. Rollers shall be constructed either of 16-gage corrosion-resistant metal or nylon, neoprene, acetal resin, or other suitable material or combination of suitable materials having adequate strength to support the loaded belt. Materials for exposed roller surfaces shall be resistant to the effects of moisture, grease, food acids, and other deleterious foodstuff ingredients. When specified, a hanging return monorail system may be used in lieu of a roller system (see 6.2).

3.10 Electrical requirements. Each counter shall be furnished complete with a motor, controls, and wiring and shall be approved for the purpose intended by a nationally recognized agency. Approval shall be evidenced by the attachment of the agency seal or label indicating the standard or specification under which the equipment was approved. Agency approval shall not absolve the contractor from compliance with the requirements of this specification. When specified, motors shall be equipped with a ground-fault circuit interrupter to protect personnel from possible electric shock injury (see 6.2).

3.10.1 Conveyor drive system. The drive section may be either a mechanical V-belt drive system or an electronically controlled dc motor drive system.

3.10.1.1 Mechanical system. The motor shall be of the horizontal type, designed, and rated for continuous duty in accordance with the applicable provisions of NEMA MG-1. The motor shall be splash-proof, and UL approved for intended use. Overload protection with manual reset and undervoltage protection shall be provided. The gear motor shall operate on 120 volts (V), single phase, 60 Hertz (Hz), alternating current (ac) system, or 220 V, single phase, 60 Hz, alternating current (ac) when specified (see 6.2). The continuous horsepower rating of the motor shall be adequate to operate the conveyor. Transmission of power to the conveyor shall be accomplished by means of a reduction gear, V-shaped belt, roller chain drive, and a sprocket that engages the endless belt. The unit shall be silent running, and an adequate chain guard shall be provided. The drive mechanism shall be designed to provide a variable belt speed from 0 to 60 feet per minute (fpm).



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3.10.1.2 Electrically controlled system. The conveyor shall be powered by a variable speed, totally enclosed, fan cooled, dc motor, through a reducing gear linkage. The motor shall be rated for continuous duty in accordance with NEMA Standard MG-1. Motor speed shall be controlled by an electronic controller which meets the applicable requirements of NEMA Standard ICS-3 and preset at the factory for minimum, and maximum speeds as specified. The motor power shall be 120 V, single phase, 60 Hz, or 220 V, single phase, 60 Hz when specified (see 6.2). The belt speed shall be adjustable over a range of 5 to 60 fpm. Provisions shall be made within the controller to provide under-voltage, overvoltage and motor overload protection. Motor overload protection shall be in accordance with NFPA NEC article 430. Wiring shall be in accordance with NFPA National Electrical Code and all dc control wiring shall be run in shielded cable. The controller shall be encased within a watertight enclosure in accordance with NEMA Standard ICS 6-110.15 type 4.

3.10.2 Controls. Each conveyor counter shall be equipped with integrally mounted controls. The controls shall include a start-and stop push button, or on/off selector switch (two-position, UL listed, rotatable switch), conveniently located for use by the operator. Controls shall also include two limit switches located near the discharge end of the conveyor. The limit switches shall stop the flow of electric current to the motor when two or more trays accumulate. When one tray is removed, the conveyor shall automatically restart. When the counter is tested as specified in 4.6.1 and 4.6.3, the switches shall be positive in action and operate as intended.

3.10.3 Electrical wiring and components. All wire, cord, and electrical components such as controllers and switches shall conform to the applicable requirements of the National Electrical Code. When type II counters are specified, the counter shall be furnished with not less than 8 feet (2.44 meters (m)), type SO, 2-conductor with ground flexible cord. The cord shall terminate in a three prong male plug, conforming to NEMA WD-1. The cord and plug shall have a current rating of at least the minimum required for full load demand. Electrical power outlet plug of 110 V or 220 V, as applicable or when specified (see 6.2), shall be provided.

3.11 Type I. Type I counters shall be stationary, mounted on 1.625-inch (41.275 mm) outside diameter, 16-gage corrosion-resisting steel tubular legs and solid adjustable feet. Legs and feet shall conform to NSF No. 2, and shall be fastened to the counter to prevent accumulation of dirt. Maximum spacing between supports shall be 6 feet (1.83 m). Intermediate supports shall hold return idlers in position. Conveyor counters, except for model b head and scrapping sections shall be provided with bell-, pear-, or bullet-feet having adjustable leveling screws. The drive and scrapping sections shall have flanged footings with means for adjustment. A minimum of 6 inches (152.4 mm) unobstructed clearance shall be provided between the floor and the lowest horizontal member of the conveyor counter.

3.12 Type II. Type II mobile counters shall be provided with heavy-duty swivel, single wheel, double ball bearing casters. The wheels, including a minimum tread width of 1.357 inches (34.468 mm) shall have a diameter of 5 or 6 inches (127 or 152.4 mm), as specified (see 6.2). The tread shall be



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composed of phenolic resin, polyurethane elastomeric rubber, or polychloroprene compound. The casters shall support the conveyor counter and all components. All casters shall be provided with foot operated brakes.

3.13 Class 1. Class 1 conveyor counters shall be provided without a belt washer.

3.14 Class 2. Class 2 counters shall be provided with a belt washer installed in the head section. The washer shall consist of a tank and manifold with spray nozzles to wash both sides of the belt. The tank shall be not less than 14-gauge corrosion-resisting steel, all welded construction, and fitted with removable, basket type, perforated, corrosion-resistant steel scrap trays. The trays shall be provided with lift-out handles. The bottom of the tank shall have a 1-1/2-inch (38.1 mm) waste outlet with a threaded tailpiece and coupling. A mixing valve, check valves, and shut-off valves shall be mounted in the head section. A water regulator and gauge shall be mounted in-line to regulate pressure of incoming water prior to introduction of detergent. A wiper shall remove excess water from the belt into the tank as the belt leaves the head section. A solenoid valve shall automatically start and stop the belt washer upon starting and stopping of the belt. An automatic detergent dispenser shall also be provided. The dispenser shall be installed to provide adequate protection against back syphonage of compounds into the water lines. A suitable mechanical and electrical connection shall be provided to permit use of detergent controls when the controls are not installed by the conveyor manufacturer. The recommended detergent used shall be nontoxic, and shall contain no ingredients harmful or injurious to personnel or equipment.

3.15 Model a. Model a conveyor counters shall be provided without a garbage disposal unit, silver soak sink, and scrapping trough.

3.16 Model b. The model b counters shall be provided with a silver soak sink and trough integral with the conveyor unit. The sink shall be constructed of not less than 14-gage corrosion-resisting steel, and shall be approximately 4 feet (1.2 m) long, 8 inches (203.2 mm) wide, and 11 inches (279.4 mm) deep. The sink shall be properly supported by brackets connected to the conveyor frame, and the bottom shall pitch toward a 2-inch (50.8 mm) center drain having a hand lever drain valve. A hot-and-cold water mixing faucet shall be supplied. Overflow water from the sink shall flow into a corrosion-resistant steel trough having approximate dimensions of 8 inches (203.2 mm) deep and 5 inches (127 mm) wide. The trough shall be of the required length to allow scraps and water to flow into the scrapping well. The bottom shall slope toward the scrapping section in accordance with the manufacturer's standard practice to assure removal of all waste from the trough.

3.16.1 Transfer table. The model b conveyor counter shall be furnished with a corrosion-resistant steel transfer table, as shown on the drawing. The table shall be equipped complete with a scrapping well, waste disposal unit, leg supports and all necessary drains, fittings, and piping.

3.16.1.1 Scrapping well. The scrapping well (cut into the table) shall be made of 14-gage corrosion-resistant steel, and shall allow scraps and waste water to flow from the trough to the waste unit.



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3.16.2 Waste disposal unit. Unless otherwise specified (see 6.2), the waste disposal unit shall conform to the requirements of MIL-G-15840, except as specified herein. The unit shall be self-contained, with provision for 1/2-inch inside diameter water supply piping. The unit shall be of the vertical type, and supported on adjustable legs (feet). The garbage disposal unit should be driven by a drip-proof, self-cooled, sealed-bearing type motor having a minimum rating of 7.5 horsepower (hp). The motor shall be designed for operation on 208 V, 3 phase, 60 Hz, ac system. The motor shall be completely wired and have a magnetic starter equipped with proper overload protection of the manual reset type. A start-and-stop push button shall be conveniently mounted for use by the operator. All water piping required for operation of the unit shall be furnished.

3.16.3 Prerinse assembly. When specified (see 6.2), a built-in stainless steel prerinse assembly shall be furnished. The assembly shall have provision for connection to 1/2-inch (38.1 mm) hot-and-cold water supply piping to permit rinsing of soiled ware over the garbage disposal unit. The assembly shall include a nonwater-absorbing and heat-resistant, flexible metal-encased, natural- or synthetic-rubber or plastic hose with sprayhead, and overhead chain control. The sprayhead shall have an adjustable means to control the spray. The spray control shall be furnished with a locking device to provide a continuous water spray.

3.17 Shelves. When specified (see 6.2), one 16-gage corrosion-resistant steel drop shelf shall be provided, and two 16-gage corrosion-resisting steel shelves for mounting over the belt shall also be provided. Shelves mounted over the belt, when provided, shall be furnished with fluorescent lighting fixtures installed under the shelves. The size, location, and mode of mounting (fixed or adjustable) of the shelves shall be as specified (see 6.2).

3.18 Lubrication. Unless otherwise specified (see 6.2), means for lubricating shall be provided in accordance with the manufacturer's standard practice. Parts requiring lubrication shall be so located as to make the lubricating points easily visible and accessible. All parts requiring lubrication shall be properly lubricated before delivery.

3.18.1 Grease fittings. Grease fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions. A bright red mark shall be painted at each lubrication point.

3.19 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.



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3.20 Electromagnetic interference control. When specified (see 6.2), the equipment shall conform to the electromagnetic interference control requirements and test limits for class C3 equipment as specified in MIL-STD-461.

3.21 Identification marking. Identification shall be permanently and legibly marked directly on the dish-tray conveyor counter or on a corrosion-resistant metal plate securely attached to the counter at the source of manufacture. Identification shall include the manufacturer's model and serial number, name, and trademark to be readily identifiable to the manufacturer. This identification shall be located for easy reading on the outside of the unit.

3.22 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.23 Commercial publications. Manufacturer's standard commercial publications shall be furnished (see 6.3).

3.24 Spare and maintenance parts. Spare and maintenance parts shall be furnished as specified (see 6.2).

3.25 Workmanship.

3.25.1 Fabrication. Metal used shall be free from fractures, splits, punctures, dents, deterioration and shall not be bowed, or malformed. The metal shall have no sharp burrs, slivers or splinters.

3.25.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.25.3 Threaded fasteners. Threaded fasteners shall not be loose or missing. Threads shall not be missing, stripped, or broken.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth



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in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Standard compliance. Proof of compliance with the standard cited in 3.6 shall be made available to the contracting officer or his authorized representative. Certification specified under 3.6.1 will be accepted as evidence of compliance.

4.1.3 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. Packaging inspection (see 4.7).

4.2.1 First article inspection. The first article inspection shall be performed on one conveyor counter when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.5 and the tests of 4.6.1 through 4.6.3 and when applicable, 4.6.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.5, the tests of 4.6.1 and 4.6.3, and the packaging inspection of 4.7. This inspection shall be performed on the samples selected in accordance with 4.4.

4.3 Inspection lot. A lot shall consist of all conveyor counters of the same type, class, and model manufactured under the same conditions and offered for delivery at the same time.

4.4 Sampling. A random sample of conveyor counters, except those requiring 100 percent operational testing (see 4.6.1), shall be selected from each lot offered to the Government in accordance with the provisions set forth in



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MIL-STD-105. All units offered for delivery at one time shall be considered a lot for purpose of inspection.

4.4.1 Sampling for tests (see 4.6.1 and 6.6.1).

4.4.2 Sampling for examination (see 6.6.2).

4.5 Examination. Each sample dish-tray conveyor counter shall be examined for defects listed in table I. Each attribute within each classification of multiple defects shall constitute one defect.

TABLE I. Classification of defects.

| Classification | Defects | Requirement Paragraph |
|---|---|---|
| Critical: None defined. | | |
| Major: | | |
| 101 | Conveyor counter not type, class, or model required. | 1.2 |
| 102 | Material not as specified. | 3.4 |
| 103 | Corrosion resistant steel not as specified; surface finish not as specified, not in accordance with ASTM A480. | 3.4.1 |
| 104 | Parts, components, and assemblies not identical to permit interchangeability. | 3.5 |
| 105 | Counter does not meet the requirements of NSF No. 2 | 3.6 |
| 106 | Performance not as required; operation of counter not automatic as required. | 3.7 |
| 107 | Design not in accordance with ANSI B20.1 and other safety requirements specified; screw threads do not conform to H28. | 3.8 |
| 108 | Construction not in accordance with Navy Food Service Systems Office Drawing Project 11103; File 541, and NSF No.2, and otherwise not as specified; top of counter not formed as required, and not required height above floor; Length of counter, pan, and belt not as specified; edges not as required; bearings not type specified; component parts not complete and assembled as required; moving parts, starting and stopping and adjusting devices do not operate as required. | 3.9 |



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TABLE I. Classification of defects. (Continued)

| Classification | Defects | Requirement Paragraph |
|---|---|---|
| 109 | Drain pan not gage and width required; material not corrosion-resistant steel; pan not under full length of belt; not constructed and mounted for removal and replacement without disassembly of other components; does not pitch toward drain; and corners not in accordance with NSF No. 2. | 3.9.1 |
| 110 | Head- and tail-end sections not completely enclosed with required material; interior not provided with required access; fittings not threaded or missing; snub- and head-pulleys not mounted as required; belt take-up device inadequate or missing; shaft and pulleys not stainless steel. | 3.9.2 |
| 111 | Length and width of belt not as specified; belt not resistant to grease, food acids, etc.; not type specified; slat-type not assembled on stainless steel chain. | 3.9.3 |
| 112 | Idler rollers are not nonmarking; not diameter, gage, and material required; not of strength required to support belt; not ball-bearing type with permanent lubrication; not resistant to moisture, grease, and food acids. | 3.9.4 |
| 113 | Electrical requirements not as specified; motor, controls, and wiring not approved as required. Motors not equipped with ground-fault circuit interrupters, when specified. | 3.10 |
| 114 | Motor not horizontal type, not designed and rated for continuous duty in accordance with NEMA MG-1, not splash-proof; magnetic starter not provided with under-voltage protection; means for transmission of power not as required; chain guard missing; belt speed not as specified. | 3.10.1.1 |
| 115 | Motor not totally enclosed, fan cooled dc motor; not designed and rated for continuous duty in accordance with NEMA MG-1; electronic controller not in accordance with NEMA ICS-3; belt speed not as specified; enclosure does not meet NEMA ICS-6 type 4 standard. | 3.10.1.2 |



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TABLE I. Classification of defects. (Continued)

| Classification | Defects | Requirement Paragraph |
|---|---|---|
| 116 | Controls not integrally mounted on counter; start-and-stop push-button missing, or in an inconvenient location; number of limit switches not as specified; switches not located and not positive in action as required. | 3.10.2 |
| 117 | Electrical wiring, core, controllers, and switches do not conform to NPFA No. 70; length and type of cord, and outlets not as specified. | 3.10.3 |
| 118 | Legs for type I counters not correct diameter, gage, and material specified; legs and feet not in conformance with NSF No. 2; maximum spacing hold return idlers not as required; intermediate supports do not hold return idlers as required; feet not as specified, leveling screws missing, or not adjustable, and drive and scrapping sections not provided with flanged footings having means for adjustment; unobstructed clearance not as required. | 3.11 |
| 119 | Diameter of casters for type II counters, and tread width not as specified; casters not heavy-duty swivel, single wheel, double ball bearing; tread not material specified; foot operated brakes missing. | 3.12 |
| 120 | Belt washer not as specified for class 2 counters; manifold not provided with spray nozzles to wash both sides of belt; nozzles broken or missing; tank not of gage and material specified and not fitted with removable basket type, perforated corrosion-resistant steel scrap tray; tray lift-out handles missing or broken; waste outlet not as specified; mixing-, check-, shut-off-valves, water regulator and gauge not provided, or not mounted as required; wiper not provided, does not remove excess water from belt; | 3.14 |



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TABLE I. Classification of defects. (Continued)

| Requirement Classification | Defects | Paragraph |
|---|---|---|
| | solenoid valve does not start and stop washer as required; automatic detergent dispenser not provided as specified, not installed with suitable backflow preventor; mechanical and electrical connection not provided, as required. | |
| 121 | Silver soak sink and trough not size, material required, and not integral with counter; sink not supported as required, and bottom does not pitch toward center drain; drain not size specified, hand lever drain valve not provided; hot-and-cold water mixing faucet not furnished; no provision made for overflow water to flow from sink to trough; bottom of trough not pitched toward scrapping section. | 3.16 |
| 122 | Transfer table not of required material, not in accordance with drawing; table disposal unit, leg supports, and required drains, fittings, and piping. | 3.16.1 |
| 123 | Scrapping well not gage, material required; scraps and water do not flow to waste disposal unit as required. | 3.16.1.1 |
| 124 | Waste disposal unit not self-contained, not as specified, and otherwise not in conformance with MIL-G-15840; no provision for water supply piping; unit not vertical type, and not supported as required; motor rating, type, and design for operation on electrical power not as specified; motor wiring, magnetic starter with required overload protection not provided as specified; start-and start-switch, and water piping missing, and not as specified. | 3.16.2 |
| 125 | Prerinse assembly not of required material; water supply piping not size required; hose, sprayhead, and chain control with locking device not furnished as required. | 3.16.3 |
| 126 | Type, gage, material, size, and location of shelves not as required; type and location of lighting not as specified. | 3.17 |



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TABLE I. Classification of defects. (Continued)

| Classification | Defects | Requirement Paragraph |
|---|---|---|
| 127 | Lubrication inadequate; lubrication points not visible and inaccessible. | 3.18 |
| 128 | Grease fittings not marked, inaccessible, and do not conform to MIL-F-3541. | 3.18.1 |
| 129 | Inadequate fungus treatment of electrical connections, components, and circuit elements when specified. | 3.19 |
| 130 | Electromagnetic interference suppression not provided when specified MIL-STD-461. | 3.20 |
| 131 | Spare and maintenance parts missing. | 3.24 |
| 132 | Metal fabrication not as specified. | 3.25.1 |
| 133 | Welding not performed as required; not free of burn holes, cracks, fractures, and undercuts; incomplete fusion; scale or flux deposits not removed. | 3.25.2 |
| 134 | Threaded fasteners loose or missing; threads stripped, broken, or missing. | 3.25.3 |
| Minor: | | |
| 201 | Identification marking not as specified. | 3.21 |
| 202 | Instruction plates not as specified or missing | 3.22 |
| 203 | Commercial publications missing. | 3.23 |

4.6 Tests.

4.6.1 Operational test. The first article sample and the samples selected in 4.4 shall be completely assembled and connected to the applicable electric, water, and drainage service. The dish-tray conveyor counter shall be operated for not less than 1 hour without dishes or trays. The counter shall operate smoothly, and without undue vibration. There shall be no evidence of failure, overheating of bearings, and all controls shall operate in a positive, smooth, and efficient manner.

4.6.2 Leak test. The first article and each item of production shall be operated in accordance with the manufacturer's instructions for a minimum period of 15 minutes. When applicable, the belt washer tank and the silver soak sink shall be filled with water. Water shall be allowed to flow on the belt during the test. Water from the silver soak sink shall overflow into the trough to the scrapping well and to the waste disposal unit. The drain pan, sink, trough, drains, tanks, garbage disposal unit, valves, pipes, and fittings shall show no evidence of leaks.



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4.6.3 Performance test. The first article sample and the samples selected in 4.4.2 shall be tested to determine that the conveyor speed, limit switches, belt washer, and garbage disposal unit comply with the requirements specified in 3.10.1, 3.10.2, 3.14, and 3.16.2, respectively.

4.6.4 Electromagnetic interference control. When electromagnetic interference control is required, the first article sample, equipped for the reduction of electromagnetic interference in accordance with 3.20, shall be tested by the contractor in accordance with test methods CE01, RE03 of MIL-STD-462. The contractor shall furnish the contracting officer, for approval, the interference control plan, the EMI-EMC test plan and the test report required in MIL-STD-461. Upon approval of the test report by the contracting officer and provided all other requirements of the specification are met, the first article sample shall be used as a model for all other units.

4.7 Packaging inspection. The preservation, packing, and marking of the dish-tray conveyor counters shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-K-43875 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Dish-tray conveyor counters are intended as conveyors of trays serviced for hospital patients in galley cold food loading areas. The counters are also used within dishwashing rooms to move soiled trays, dishes, glassware, and silverware from galley mess halls to galley kitchens.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and model dish-tray conveyor counter required (see 1.2).
c. When conveyor counter is enclosed pan type (see 3.1).
d. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
e. Length of conveyor counter (see 3.9).
f. Belt width of 10 or 12 inches (254 or 304.8 mm); slat- or flexible-type belt (see 3.9.3).
g. When belt return system is hanging monorail type (see 3.9.4).
h. When ground fault circuit interrupters are required (see 3.10).
i. When 220 volt operation is required (see 3.10.1.1, 3.10.1.2, and 3.10.3).



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j. Whether caster is 5 or 6 inches (127 or 152.4 mm) in diameter (see 3.12).
k. Type of waste disposal unit/system, if other than as specified (see 3.16.2).
l. When a prerinse assembly is required (see 3.16.3).
m. When shelves are required, the number, size, location, and mode of mounting (fixed or adjustable) (see 3.17).
n. Whether means for lubrication is not required (see 3.18).
o. When fungus resistance treatment is required (see 3.19).
p. When electromagnetic interference suppression is required (see 3.20).
q. Spare and maintenance parts required (see 3.24).
r. Level of preservation and packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete conveyor counter or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Part or Identifying Number (PIN). The PIN to be used for dish-tray conveyor counters acquired to this specification are created as follows:

| M | 23565- | XX | X | X |
|---|---|---|---|---|
| Prefix to indicate military specification | Specification number | Type number (see 1.2) | Class number (see 1.2) | Model letter (see 1.2) |

6.6 Sampling procedures.

6.6.1 Sampling for tests. Recommended Inspection Level is S-2 and Acceptable Quality Level is 2.5 defects per hundred units (see 4.4 and 4.4.1).

6.6.2 Sampling for examination. Recommended Inspection Level is II and Acceptable Quality Level is 2.5 for major defects and 4.0 for minor defects expressed in terms of defects per hundred units (see 4.4 and 4.4.2).



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6.7 Subject term (keyword) listing.

Washer
Transport
Culinary

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - YD
Air Force - 99

Review activities:
Army - MD
Navy - SA, MC
Air Force - 84

User activities:
Army - CE
Navy - MS

Preparing activity:
Navy - YD

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